Weekly Lesson Plan — Week 10 · Semester 1 · 2026–27 · Da Vinci School

Teacher: Gavaldon | Week of: October 12, 2026 | Course: Engineering Design & Problem Solving (InSPIRESS)

TEKS / ELPS:

TEKS: §127.406(6) implement a design process to develop a multi-system product or solution; 127.406(4) communicate findings using appropriate tools
ELPS: c.1C, c.2C, c.3E, c.4C, c.4F, c.5G

Aim of the Lesson / Objective: *Statement of what you want students to do (skill) and/or know (knowledge). Students will…*

MONDAY OCT 12	TUESDAY OCT 13	WEDNESDAY OCT 14	THURSDAY OCT 15	FRIDAY OCT 16
NO SCHOOL	Students will construct a midplane, model a hex nut, and apply a modeled thread with the correct direction and size. **ALTERNATE:** Students will write 3–4 paragraphs analyzing how automation and artificial intelligence change employment, and post photographic evidence of the day's work.	Students will define the CAD tools used so far in their own words and sketch a part by hand, labelling the features used to build it. **ALTERNATE:** Students will write 3–4 paragraphs analyzing how automation and artificial intelligence change employment, and post photographic evidence of the day's work.	Students will finish the hex nut and begin the handlebar grip by embossing text onto a curved face. **ALTERNATE:** Students will write 3–4 paragraphs analyzing how automation and artificial intelligence change employment, and post photographic evidence of the day's work.	Students will complete the handlebar grip using a body pattern, an extrude to an object, a parametric shell, and a combine.

Employed Language Domain(s) Objective: *How students achieve the objective – Listening, Speaking, Reading, Writing.*

Students will collaborate in teams to investigate engineering problems, communicate their findings, and present their solutions. They listen to team members' ideas, read technical directions, and write explanations of their designs. Wednesday is a writing- and sketching-heavy day with no computers. Alternate-track students read, write and revise a multi-paragraph study of automation and employment.

Assessment of Content Objective: *Instrument used to determine mastery.*

Main project: the model file in each student's own cloud folder, checked against the day's feature list and the timeline. Wednesday: the completed CAD crossword and a labelled hand sketch. Alternate assignment (Tue–Thu): a daily photo of 3–4 written paragraphs posted to Google Classroom.

Set (Bell Work): *A motivating action that engages students in the objective. Essential Question:*

MONDAY OCT 12	TUESDAY OCT 13	WEDNESDAY OCT 14	THURSDAY OCT 15	FRIDAY OCT 16
NO SCHOOL	a bolt and a nut are made in two different factories and still fit - how?	pick any object on your desk - which three Fusion tools would build it?	why is the text on a tire or a bottle raised instead of printed?	a grip has rings and a hollow core - name one reason for each

Methodology / Steps: *What it takes to teach the objective so 100% can pass the assessment.*

MONDAY OCT 12	TUESDAY OCT 13	WEDNESDAY OCT 14	THURSDAY OCT 15	FRIDAY OCT 16
NO SCHOOL	Step 1: Bell ringer / essential question. Step 2: Bell work: a bolt and a nut are made in two different factories and still fit - how? (10 min) Step 3: Day 6 video: hex profile, midplane construction plane, extrude, Thread tool (25 min) Step 4: Chamfers: the regular chamfer and the custom one Kevin builds (8 min) Step 5: Save (Ctrl+S) as Day 6 - Hex Nut - Lastname (2 min) **ALTERNATE ASSIGNMENT:** Automation Study Day 1: what machines already replaced - 3 to 4 paragraphs + a photo posted to Classroom.	Step 1: Bell ringer / essential question. Step 2: Bell work: pick any object on your desk - which three Fusion tools would build it? (10 min) Step 3: CAD crossword: the tools from Days 1-7 (20 min) Step 4: Hand sketch: design a part with your name embossed on it, label the features (13 min) Step 5: Turn in both pages (2 min) **ALTERNATE ASSIGNMENT:** Automation Study Day 2: the company's side - cost, speed, quality - 3 to 4 paragraphs + a photo posted.	Step 1: Bell ringer / essential question. Step 2: Bell work: why is the text on a tire or a bottle raised instead of printed? (10 min) Step 3: Finish Day 6: chamfers and mirror; save (8 min) Step 4: Day 7 video: base cylinder, 3D text, Intersect to centre it, Emboss (25 min) Step 5: Save (Ctrl+S) as Day 7 - Grip - Lastname (2 min) **ALTERNATE ASSIGNMENT:** Automation Study Day 3: the worker's side - who loses the job and what retraining costs - 3 to 4 paragraphs + a photo posted.	Step 1: Bell ringer / essential question. Step 2: Bell work: a grip has rings and a hollow core - name one reason for each (10 min) Step 3: Day 7 video: offset rings, rectangular pattern of bodies, extrude To Object, shell, combine (28 min) Step 4: Folder check: Day 6 and Day 7 both saved in Students > your name (5 min) Step 5: Save (Ctrl+S) (2 min)

Resources: *Textbook, handouts, visuals, laptops.*

Google Classroom, projector, laptops, Autodesk Fusion, the "Learn Autodesk Fusion in 30 Days" 2026 Edition videos (Days 6 and 7), engineering notebook, printed CAD crossword, alternate-assignment handout.

Re-Teaching Activities: *What you prepare if students do not master the objective.*

Rewind-and-rebuild at the teacher table for the feature that failed; printed step list; one-on-one at the student's screen; peer re-teach inside the seating group. Alternate track: sentence stems and a conference.

Reflection: *Students analyze what they accomplished and how they reached mastery.*

Exit ticket – students state which feature they finished and the one mistake they fixed. Wednesday: students name the three tools that would build the object they sketched.

Study Skills: *Skills implemented by students.*

Note-taking, technical vocabulary, following multi-step directions, self-checking against a list, teamwork, written argument.

SPED & 504 Accommodations / Modifications: *Codes.*

Per Individualized Education Plan (IEP) / 504 plan.

ELL Accommodations: *Codes.*

Clarify directions (CD); Extra time (ET); Peer and Native Language Support (PN); Pre-teach vocabulary (PV); Rephrase, Repeat, Slow Down (RS); Visual Cues (VC); Wait time (WT); Pictorial Representation (DP). GROUPING: mixed-proficiency small groups so emergent bilingual students hear and use the academic vocabulary in context and rehearse with a peer before whole-class response (PN, WT). ONE-ON-ONE: individual teacher explanation on request every period; directions restated and modeled at the student desk (CD, RS, VC).

Notes: *Schedule changes this week.*

Monday Oct 12 – no school (district calendar). Wednesday Oct 14 – SAT testing displaces the room and no laptop cart is available; the day runs on paper. Friday Oct 16 – Mr. Gavaldon leaves at 10:00 AM with the chess team for the RGPI tournament; later periods follow the Week 10 Classroom post independently.